measurement of the sorption of radiotracers which do not penetrate into the specimen.

If a radioactive gas is incorporated in a crystalline compound the amount of gas released (the *emanation ability*) can be measured as a function of the temperature. It is found that the emanation increases considerably at certain temperatures, indicating structural changes in the solid at those temperatures. Studies of diffusion and emanation play a valuable role in understanding the mechanism of sintering and in the formation of new solid compounds. This has been of practical importance in the cement and glass industries, in the production of semiconductors, in the paint industry, etc. Studies of surface reactions are of practical importance for flotation, corrosion, metal plating and finishing, and detergent action to name only a few applications.

The distribution of a radioactive element or compound in a composite matrix can made visible either to the naked eye or under a microscope by means of *autoradiography*. The technique is based on the blackening of photographic films when exposed to nuclear radiation (cf. §§8.2 and 8.10). The technique is best illustrated by an example.

Lead is an unwanted impurity in stainless steel even in very small amounts. In order to investigate the mechanism of its incorporation, ^{212}Pb was added to a steel melt. After cooling, the ingot was cut by a saw and the flat surface machine-polished and etched in an electrolytic bath (electro-polishing). This provided a very flat and "virgin" surface. A photographic film was placed firmly with even pressure against the metal surface, and the film was exposed in darkness in a cool room for about a week. After development of the film, darkened spots caused by the radiation from ^{212}Pb showed where on the metal surface lead was present. By taking the results into account in the production process, the negative effect of lead in the raw material could be reduced.

In another technique, a polished surface of the specimen (metal, mineral, etc.) was dipped into a solution containing a radioactive reagent, which selectively reacts with one of the constituents of the surface. A mineral was dipped into potassium ethyl xanthate labelled with ^{35}S (β^- 87.5 d); the xanthate reacted selectively with sphalerite (zinc blende), ZnS, in the sample. The distribution of the xanthate, as shown by the autoradiograph, indicated the ZnS distribution in the mineral. The low β-energy of ^{35}S, E_{max} 0.2 MeV, was an advantage to the technique because the resolution of the autoradiograph increases with decreasing particle range.

18.5. APPLICATIONS TO LIFE SCIENCES

The largest field of application of radionuclides is in the life sciences. A survey is presented in Table 18.3.

In the reactions leading to the desired product (column A) many factors must be considered. They are of such importance that we devote several separate chapters to this: Ch. 10 on nuclear reactions, Ch. 16 on particle accelerators and Ch. 17 on production of

Table 18.3 Survey of radionuclide use in life sciences

A Radionuclide production
- **1** Target chemistry,
- **2** irradiation,
- **3** isolation
- **4** and processing, yields
  - **(a)** pure radionuclide, or
  - **(b)** labelled compound (radiopharmaceuticals)

B Source, position/ administration
- **1** Internally (α,β or γ)
  - **(a)** injection,
  - **(b)** inhalation, or
  - **(c)** oral intake.
- **2** Externally as γ-source

C Technique/application [detection]
- **1** Biochemical analysis
  - **(a)** autoradiography [photographic]
  - **(b)** immunoassay [counting]
  - **(c)** DNA-analysis [photographic]
  - **(d)** direct tracing [counting]
- **2** Medical imaging
  - **2.1** Transmission Tomography (TCT) photographic, or by counting → computer → display]
  - **2.2** Emission Computed Tomography (ECT) [counting → computer → display]
    - **(a)** Single Photon Emission Computed Tomography (SPECT)
    - **(b)** Positron Emission Tomography (PET)
- **3** Irradiation uses
  - **(a)** by internal sources (therapeutically)
  - **(b)** by external sources (therapeutically)
  - **(c)** by external sources (sterilization etc)

radionuclides. The incorporation of the radionuclide in a chemical compound (labelling, §17.5.3) provides it with unique properties, such as specific biological affinity (§18.5.1). When such labelled compounds are taken up by organisms (A4b → B1) they move to specific sites in the organs, signalling normal or abnormal behavior. When used in medicine (primarily C1 and C3) these compounds are referred to as *radiopharmaceuticals* (A4b).

The use of labelled compounds in life sciences is extensive, in fact, the largest single user of radionuclides is medical science. It has been said that radioactive tracers have been of equal importance to medicine as the discovery of the microscope. Presently one out of ten hospitalized patients in the United States is admitted to some nuclear medical procedure. If the intended use of the radionuclide is as an external radiation source (A4a → B2 → C2.1 or C3) its chemical matrix is of minor importance. Such sources are used for radiation treatment of cancer (C3b), radiation sterilization of food (C3c), etc. The *radiation effects* on biological systems are discussed separately in Ch. 15. In this chapter we focus our interest on radionuclides with specific chemical properties, in the order of column C, Table 18.3.

18.5.1. Biological affinity

To study how living species interact with the environment, *ecology*, one can use radioactive tracers to follow the uptake of a trace metal (e.g. cobalt) from the soil by

plants, and by animals after having eaten the plant. In agriculture, this is useful in studying the uptake of trace elements necessary for plant growth. For example, it has been found that sheep need plants containing selenium in order to combat white muscle disease. The turnover in nutrients fed to animals can be determined; it was found that 20% of the phosphorus in cow's milk comes directly from the feed, while 80% is taken from the cow's bone.

If a radioactively labelled compound such as an amino acid, a vitamin, or a drug is administered to an animal, the substance is incorporated to varying degrees in different organs (*biological affinity*). The substance undergoes chemical exchange with other substances in the body, is broken down, and, finally, discharged from the body (i.e. *metabolizes*). The radionuclide distribution in samples (cells, tissues, body fluids, etc) removed from living species gives significant information about the (normal or abnormal) physiology of that species. "C1. Biochemical analysis" in Table 18.3 lists these techniques, which are discussed below.

(a) *Autoradiography*

Figure 18.12(a) shows the distribution of ^{14}C-labelled p-aminosalicylic acid, PAS, the first tuberculostatic agent developed, in a 20 μm thick section of a mouse. The autoradiograph was obtained after 10 MBq ^{14}C-PAS had been injected, and the mouse (weight 20 g) had been killed by immersion into a CO_2-acetone (−80°C) mixture and sectioned. It is seen that the PAS is concentrated mainly in the lung, where it is effective against tuberculosis, and the kidney and intestine as it is excreted through these organs.

Figure 18.12(b) shows an autoradiograph of radioactively labelled cells. In the "pulse labelling" (i.e. 10 − 30 min exposure) to ^{3}H-thymidine only cells in the S-phase of the cell cycle reacts with the thymidine and thus become labelled (the very dark spots). Both the cell cycle times and number of cells in the cycle can be measured with this technique, which is important in cancer cell research.

As an alternative to using the blackening of a photographic film for radiation detection "instant imagers" based on semiconductor array detectors are commercially available.

(b) *Radioimmunoassay (RIA)*

Immunoassay is an application of the substoichiometric principle (§18.3.4) developed by Yalow (Nobel laureate in 1977) for protein analysis. In the United States tens of millions radioimmunoassays are made annually in hospitals to measure hormones, enzymes, viruses, serum proteins, drugs, and so forth. Only a drop of the patient's blood is needed, reflecting the versatility and sensitivity of this technique, which can be performed automatically. Commercial RAST-kits (Radio Allergy Sorbent Tests) are used for rapid diagnosis of allergic reactions.

In immunoassay, a known mass, w_s^0, of a labelled protein P* is allowed to react with a much smaller (substoichiometric) mass of an antibody A, so that a complex P*A is formed. The P*A is isolated and its radioactivity, R_s, measured. Under the same

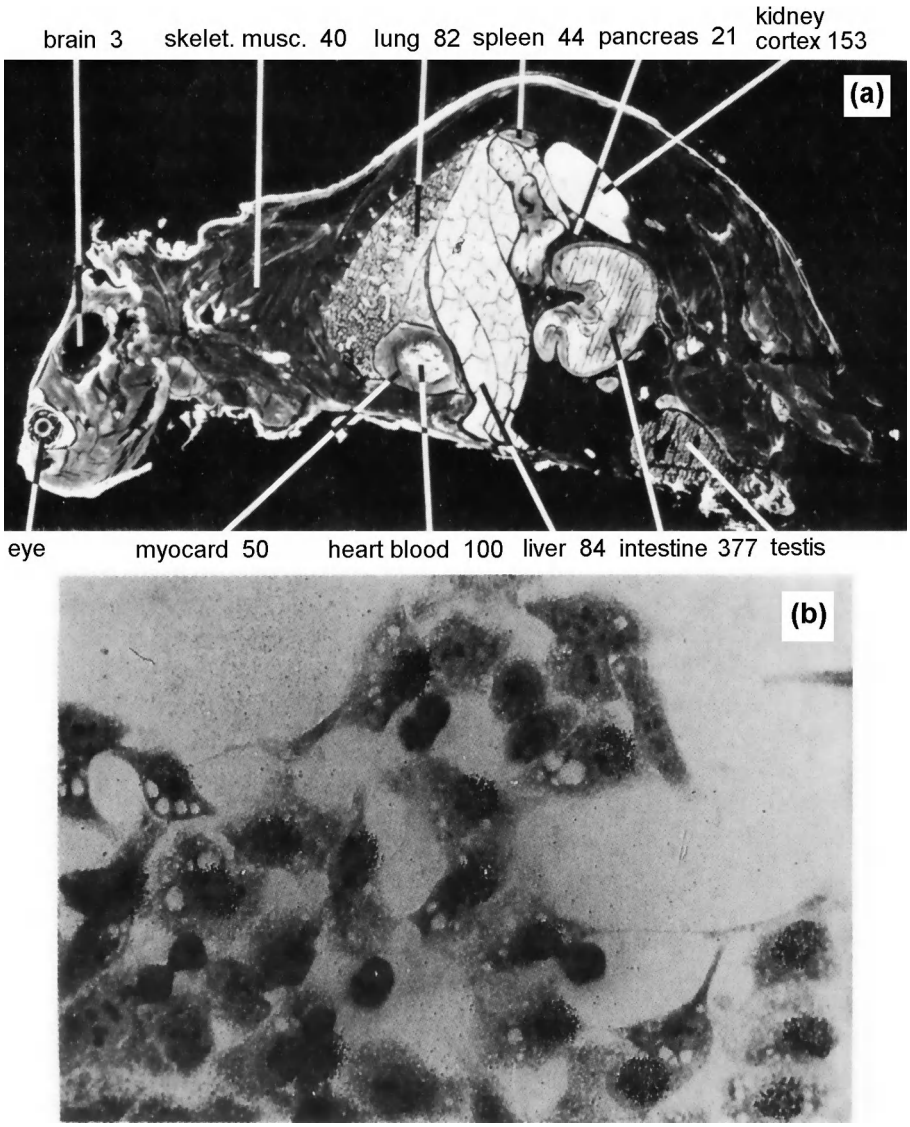


Figure 18.12 Autoradiograms, (a) distribution of ^{14}C-PAS in a mouse 30 minutes after intra-venous injection, (b) of HeLa-cells labeled by ^{3}H-thymidine. *(From Hanngren and Nias.)*

conditions, an identical mass, w_s^0, of labelled protein is mixed with an unknown mass, w_u, of the identical protein to be determined. This sample is also allowed to react with the same amount of antibody A as before; the complex P*A is again isolated (weight w_{u+s}) and its radioactivity, R_{u+s}, measured. The unknown weight, w_u, is then calculated from (18.17).

(c) *DNA-analysis*

The chemical composition (the base sequence) of the DNA molecules, which make up the chromosomes of the cell nucleus, is unique for each species and individual. The detailed analysis of the DNA molecule provides important information about its host. This is used in studies of the evolution of species, in forensic science to identify criminals (e.g. from blood, sperm, etc), in transplantation of organs (kidney, tissue grafting, etc), in detection of genetic diseases, etc. The importance of DNA-analysis is constantly growing.

The sequence of the DNA analysis is as follows. (i) The cell walls are broken up by osmosis etc. and the doublestranded DNA is denatured to pieces of singlestranded DNA. The molecules may be concentrated by centrifugation. (ii) By exposing this DNA to *restriction enzymes* the DNA nucleotide chain is sectioned further into smaller fragments; different restriction enzymes cut the DNA molecule at different positions. The fragments are exposed to radioactively labelled compounds (e.g. containing ^{14}C or ^{32}P) which selectively attach to the different fragments. Alternatively to (i) and (ii), the DNA is directly labelled e.g. by ^{32}P in a cloning process; the clones will then contain ^{32}P at its P- sites also after it has been split by restriction enzymes. (iii) When the DNA samples are exposed to electrophoresis in a suitable gel (agarose, polyacrylamide, etc), the fragments distribute themselves along the potential direction according to their migration velocities. Usually 10—40 samples (each treated by a different restriction enzyme) are run simultaneously. (iv) After the electrophoresis, a photographic plate is placed on the gel and exposed to the radioactive fragments, producing, after development, a pattern of spots or bands on the plate. (v) The band pattern is analyzed to reveal its information about the individual. The technique has many similarities with those demonstrated in Figures 18.7 and 18.8.

18.5.2. Transmission computer tomography (TCT)

Conventional "x-ray pictures" show the morphological structure of the internal organs. The technique is used with x-ray or γ-ray sources; e.g. Ra was extensively used during World War I in operations at the front where electricity was lacking. To improve the contrast in the photographic pictures, dense media like barium sulfate or iodine compounds are administered to the patient; this method is said to be *invasive* and can be painful to the patient.

Though this technique is still widely in use, the 1970'ies saw a large step forward in medical imaging referred to as *computerized tomography* (CT), developed by Cormack and Hounsfield (Nobel laureates in 1979) and others. Though the radiation source is the same, the photographic plate is replaced by one or several radiation sensitive detectors, a digital computer and a display. The radiation source and detector array are moved (*scanned*) in relation to the patient, see Figure 18.13 A. This technique requires computer

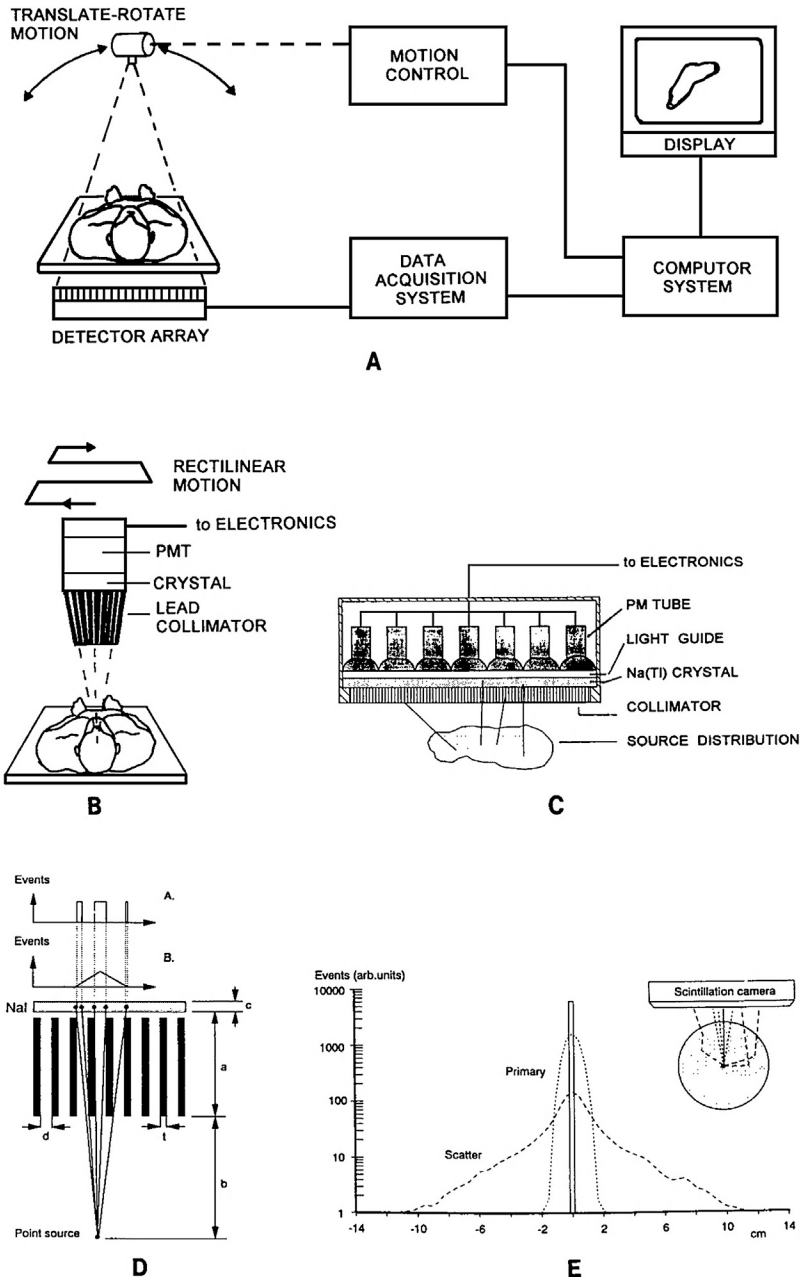


Figure 18.13 Principles of gamma scintigraphy.

software to handle the reconstruction arithmetic needed to provide an image on the screen from the observed changes in count rates by the detectors (due to different absorbancies of the organs) at different geometries. Tomography means "slice"; the technique shows slices through the body on the screen. The resolution of a TCT-scan is usually ~1 mm, allowing quite exact pictures, e.g. of a tumor. Since nowadays only X-ray tubes are used as radiation source, the technique is handled by *radiologists* at the hospital "X-ray department".

Transmission tomography can be compared to a silent movie: one can see what happens physically with the organs. Emission tomography (next section) uses internally administered radioactive isotopes, which provides explanation to why the organs function as they do. Thus, to extend the comparison, computerized emission tomography becomes more like a medical sound movie of the patient.

18.5.3. Emission computer tomography (ECT) and diagnosis

The rate of incorporation and discharge of radioactively labelled substances in the body provides a measure of the metabolism of healthy and of sick tissues. On medical patients this information is obtained by external measurements referred to as *radioisotope scanning* (RIS). Such scanning can yield information about a medical disorder much before it is observed by other means. Since the amount of radioactive tracer is very small, this technique is referred to as *non-invasive*. In hospitals the department of *nuclear medicine* is normally responsible for these investigations.

(a) *Simple scanners*

Simple scanners are designed either with one or several direction sensitive (focusing) detectors, which are moved around or above the patient in a pattern; Figure 18.13.B and C. Figure 18.14 shows the result of a kidney scan, *renography*, of a 38 year old woman who

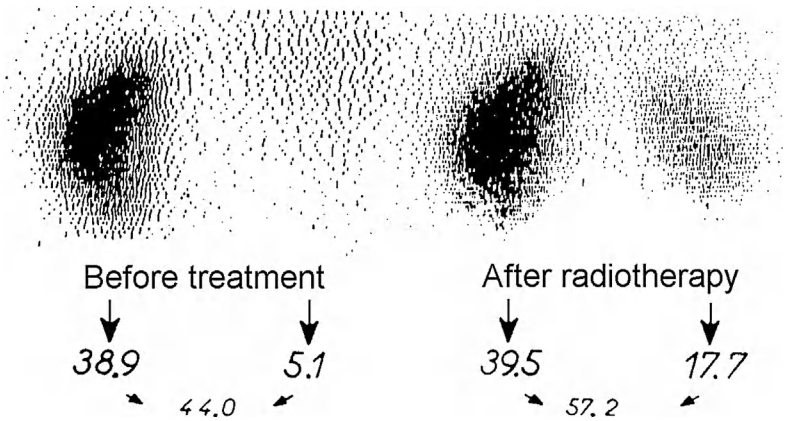


Figure 18.14 Kidney function test by $^{197}HgCl_2$ using scintigraphy. *(From Kellershohn et al.)*

has been administered $^{197}HgCl_2$ (ECγ $t_{½}$ 2.672 d); in this case a single NaI(Tl)-PMT with collimator was moved in a recti-linear pattern over the kidneys (Fig. 18.13.B). The left picture shows that 38.9% of the compound has been fixed to the left kidney, and very little to the right kidney. This was caused by a vaginal fibrous sarcoma blocking the urethra from the right kidney. After radiation therapy some improvement is seen (right Figure). Renography is now done with ^{131}I labelled hippuric acid, or a ^{99m}Tc complex.

(b) *Gamma Camera and Single Photon Emission Computer Tomography (SPECT)*

Recent advances in nuclear medical imaging are the gamma camera, SPECT and positron emission tomography (PET, described under (c)).

Figure 18.13.C shows the gamma scintillation camera, *gamma camera*, originally developed by Anger. It consists of a two-dimensional array of 40–100 PMTs (often hexagonally formed for tight stacking) viewing a large flat NaI(Tl) crystal of ~400 mm diameter and 5–10 mm thick, which is located behind a lead collimator containing numerous holes. Typical hole size is 2–3 mm diameter and 40 mm length. Collimator dimensions (Fig. 18.13.D) depend on the E_γ; for ^{99m}Tc (E_γ 0.14 MeV) the wall thickness is 0.2–0.3 mm.

The radionuclide injected into the patient should preferably decay by the emission of a single γ, and the lower the E_γ the better becomes the resolution of the image. Since the detectors are energy sensitive, the camera can be tuned to the primary unperturbed (Ch. 7) γ's, i.e. the signals from the PMTs are accepted only if they are near the photo-peak energy. The energy window of the detector has some width allowing some scattered photons to contribute to the number of events registered from a single point radionuclide source; this is suppressed by narrowing the energy window (Fig. 18.13 E). Figure 18.13 D indicates, in the upper diagram (A), the actual events detected through the pin holes, and how these can be sharpened into a peak if the collimator moves during the exposure time (Fig. 18.13 B); this is of particular importance in SPECT-investigations. The gamma camera is supported by hardware (Fig. 18.13.A) and extensive computer software.

One distinguishes between "scintigraphic" and "SPECT" investigations. Gamma cameras are used in both investigations, but the camera is stationary in the former case, while it usually moves in SPECT. About 95% of all nuclear imaging investigations carried out by the nuclear medicine department of a modern hospital is scintigraphy with stationary gamma camera, usually referred to as "gamma camera (investigation)". In about 2/3 of these investigations a single picture is taken of the patient's organ: heart, kidney, liver, etc. This yields an image of the organ's content of the radionuclide, i.e. of its function with respect to the pharmaceutical administered. About 1/3 of the investigations are studies of the dynamic behavior of an organ. Taking thousands of "pictures" (commonly 10 per second), storing and treating them allows a direct study of the function, e.g. heart beat. Though the resolution in such investigations is rarely better than 10 mm, they nevertheless provide information not available by TCT.

Each picture by the camera is two-dimensional; however, by positioning the camera at different angles around the patient, 3-dimensional pictures of the organs can be constructed. This is commonly done by letting the camera rotate around the patient, usually 360° in increments of a few degrees. Several large cameras at fixed positions or rotating around the patient can be used for the same purpose. The cameras can also be moved in a direction parallel to the patient. By using software similar to that used in TCT-scanning, "radiographic slices" of the patient's organs can be obtained. Such investigations are referred to as SPECT. The resolution of present commercial SPECT equipment is only 12–15 mm.

With the increasing number of radiopharmaceuticals with specific biological affinities, gamma camera and SPECT have become important diagnostic tools with numerous clinical applications, and virtually every organ in the body has been studied. Table 18.4 shows the most frequently performed imaging investigations; Table 18.5 lists data for radionuclides applied in medicine, including the amount of radionuclide needed. The radiation doses received by the patient in diagnostic investigations is usually <10 mGy per investigation.

The SPECT technique is primarily used for cardiovascular and brain imaging. Cardiac stress tests, using $^{201}Tl^{+}$ or ^{99m}Tc labelled radiopharmaceuticals, amount in the U.S. to about 2 million a year. Brain tumors can be located by SPECT after intravenous injection of $Na^{99m}TcO_4$, as brain tumors have a very high affinity for and slow release of Tc. In comparison, the uptake of Tc in brain infarcts is low and the release fast, and from healthy parts of the brain even faster; thus various constrictions to the cerebral blood flow are easily located. A head scan can be made in 10 minutes and virtually instantaneously produces an image of the brain.

Mental disorders are diagnosed by SPECT, gamma-camera or PET using various radiopharma-ceuticals, e.g. after the injection of ^{99m}Tc-HMPAO (Hexa Methyl

Table 18.4 Ten of the most frequently performed imaging investigations

Examination	Radionuclide	Static/dynamic	Principal application
Bone	^{99m}Tc	S	Secondary spread of malignancy
Liver	^{99m}Tc	S	Secondary spread of malignancy; cirrhotic changes
Brain	^{99m}Tc	S/D	Occult metastases; brain damage; vascular problems
Lung	^{99m}Tc	S	Pulmonary embolism (perfusion)
Lung	$^{133}Xe/^{99m}Tc$	D	Pulmonary emphysema (ventilation)
Kidneys	$^{123}I/^{99m}Tc$	D	Renal function
Hepato-biliary	^{99m}Tc	D	Patency of the biliary tree; liver function system
Thyroid	$^{123}I/^{99m}Tc$	S	Thyroid function
Heart	^{201}Tl	S	Cardiac infarction and ischaemia (perfusion)
Heart	^{99m}Tc	D	Cardiac wall motion (blood pool)

(From Sharp et al.)

Table 18.5 Radionuclides applied in nuclear medicine

Nuclide	Decay		Particle energy (keV)*	Applications**
	Half-life	Mode		
^{3}H	12.33 y	β^-	18.6 β	AN. BA all chem. comp.
^{11}C	20.39 m	EC β^+	960 β, 511 γ	As CO_2, HCN, HCHO, CH_3I. DG PET: brain, glucose
^{14}C	5730 y	β^-	156 β	AN
^{13}N	9.965 m	β^+	1190 β, 511 γ	As NH_3. DG PET
^{15}O	2.037 m	β^+	1723 β, 511 γ	As H_2O. DG PET
^{18}F	1.830 h	β^+ EC	635 β, 511 γ	BA skeleton. As ion or F_2. DG PET
^{24}Na	14.659 h	β^- γ	1389 β, 1369, 2754 γ	AN. BA circulatory system.
^{32}P	14.282 d	β^-	1710 β	AN. BA skeleton, bone marrow. TP: leukemia 200 MBq phosphate.
^{35}S	87.5 d	β^-	1674 β	AN
^{42}K	12.360 h	β^- γ	3523, 1970 β, 1525 γ	AN liquid volume. BA muscles.
^{47}Ca	4.536 d	β^- γ	1981, 684 β, 1297 γ	BA skeleton. DG: bone
^{51}Cr	27.704 d	EC γ	320 γ	BA spleen. DG: kidney funct. (clearance) 4 MBq EDTA.
^{55}Fe	2.73 y	EC		BA: red blood cells
^{59}Fe	44.50 d	β^- γ	475, 273 β, 1099, 1292 γ	DG: ion
^{57}Co	271.77 d	EC γ	122, 137 γ	BA liver.
^{58}Co	70.92 d	β^+	811, 511 γ	DG: vitamin B12
^{60}Co	5.271 y	β^-	315 β, 1333, 1173 γ	TP: cancer. Source $>10^7$ MBq.
^{68}Ga	1.135 h	β^+ EC	1830 β, 511, 1077 γ	BA intestine. DG: ion, EDTA or citrate.
^{75}Se	119.77 d	EC	(864) 136, 265, 280 γ	BA kidney, liver. DG: methionine
^{85}Sr	64.84 d	EC	514 γ	BA skeleton. DG: skeleton 2 MBq ion.
^{99m}Tc	6.006 h	IT	140 γ	DG: Thyroid 100 MBq TcO_4; heart 400 MBq TcO_4; lung (perfusion, emboli) 80 MBq albumin particles; liver (tumor, size, funct.) 100 MBq colloid, or 150 MBq HIDA; kidney 200 MBq DTPA (clearance), DMSA (renography), and other compounds; skeleton (metastasis) 400 MBq phosphate; brain (tumor, cerebral hemorrhage) TcO_4 or DTPA. Etc.
^{111}In	2.807 d	EC	(864) 171, 245 γ	DG: spinal cord ion, oxime or DTPA.
^{113m}In	1.658 h	IT	(392) 392 γ	DG: ion, oxime or DTPA.
^{123}I	13.2 h	EC	(1232) 159 γ	BA thyroid. DG: Kidney funct. 40 MBq hippuran; thyroid.
^{125}I	60.1 d	EC	(178) 35 γ	BA thyroid. DG: Thrombosis 4 MBq fibrinogen. Kidney function hippuran

Table 18.5 Radionuclides applied in nuclear medicine—cont'd

Nuclide	Decay		Particle energy (keV)*	Applications**
	Half-life	Mode		
^{131}I	8.040 d	β^-	606 β, 365 γ	DG thyroid funct. 2 MBq ion. Kidney 1-10 MBq hippuran. TP: 300 MBq ion hyperthyroidism and thyroid cancer.
^{133}Xe	5.24 d	β^-	346 β, 81 γ	DG Lung emboli, 1000 MBq gas (radiospirometry), brain.
^{137}Cs	30.0 y	β^-	514 β, 662 γ	TP: cancer. Source $>10^6$ MBq.
^{186}Re	3.777 d	β^-	309, 362 β, 137 γ	bone cancer
^{188}Re	16.98 h	β^-	728, 795 β, 155 γ	pain
^{198}Au	2.6935 d	β^-	961 β, 412 γ	TP: 150 MBq spread of ovarian cancer.
^{201}Tl	3.05 d	EC	(480) 167 γ	BA: para-thyroid, kidney. DG Heart (infarct) 50 MBq ion

AN = biochemical analysis, BA = biological affinity, DG = diagnostically (imaging), TP = herapeutically.
* Particle energy (decay energy)
** Common radiochemical species; ion = ionic as chloride, sulfate, etc.

Propylene Amine Oxide) or inhalation of ^{133}Xe. Injecting ~1000 MBq Tc-complex into the blood stream, about 5% of this compound moves to the brain, passes the membrane of the blood vessels and enters into the brain tissue, where it decomposes and decays with its 6.0 h half-life, allowing detailed imaging of the brain. The average absorbed whole body dose in such a SPECT scan is about 0.01 Sv. Figure 18.16 (right) shows three ^{133}Xe gamma-camera pictures of patients with Alzheimers disease (ALZ), frontal lobe dements (FLD) and multi-infarct dements (MID). ^{133}Xe decays by emitting an 80 keV γ, thus the pictures show primarily the ^{133}Xe in the brain cortex (regional-Cerebral Bloods Flow test, r-CBF); in this investigation 256 small scintillation detectors were positioned like a helmet around the patient's head. The pictures in black-and-white shown here do not make justice to the original, more detailed, color pictures, the colors representing the radiation intensity, i.e. blood flow at the point.

^{99m}Tc is a preferred radionuclide due to its convenience of production (from milking ^{99}Mo, §5.16, the ^{99}Mo produced by fission of ^{235}U), and short half-life which reduces radiation risks. There are >20 differently labelled Tc-compounds commercially available for diagnostic purposes. ^{99m}Tc (together with ^{123,125,131}I) is the most frequently used radionuclide for diagnostics; about 7 million such investigations are made per year in U.S. The dominating organs investigated are in order skeleton, kidney, liver and thyroid.

(c) *Positron emission tomography (PET)*

The decay of a positron emitting radionuclide yields two 0.51 MeV γ-rays travelling in opposite directions. If photons with this energy are registered simultaneously by γ-ray

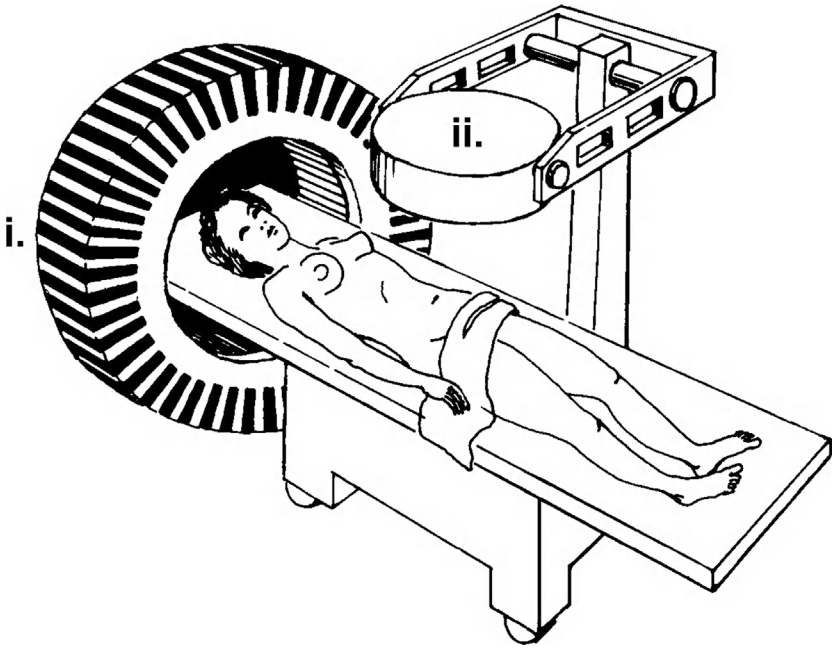


Figure 18.15 Geometry in (i) positron emission tomography (PET) and (ii) gamma-camera.

detectors 180° apart, positron decay must have taken place somewhere along the line between the two detectors. This is used for *positron emission tomography*. A positron emitter is administered to the patient positioned inside a ring or hexagon of 50—100 scintillation-PMT detectors (there is no need for collimators); Figure 18.15. The ring is moved in a translate-rotate pattern. The location of the radioisotope in the body is mapped in a way similar to that described for SPECT. The resolution of this technique is presently of the order of a few mm.

Positron emitters cannot be produced by n-irradiation: from Figure 5.8 it is seen that only charged particle irradiation (using ^{1}H, ^{2}H, ^{4}He, etc) can result in product nuclei on the proton rich side of the stability valley, for which positron emission is the main decay mode (competing with electron capture). Typical positron emitting nuclides used in PET are included in Table 18.5. They can be tagged to a variety of compounds.

For studying brain metabolism, ^{11}C-labelled glucose has been extensively used. The procedure is as follows:

1) $H_3\,{}^{11}BO_3$ is irradiated by protons in an accelerator, yielding $^{11}CO_2$.

2) Rapid automatic synthesis produces $^{11}C_6H_{12}O_6$ (glucose) or the methyl glucose derivative.

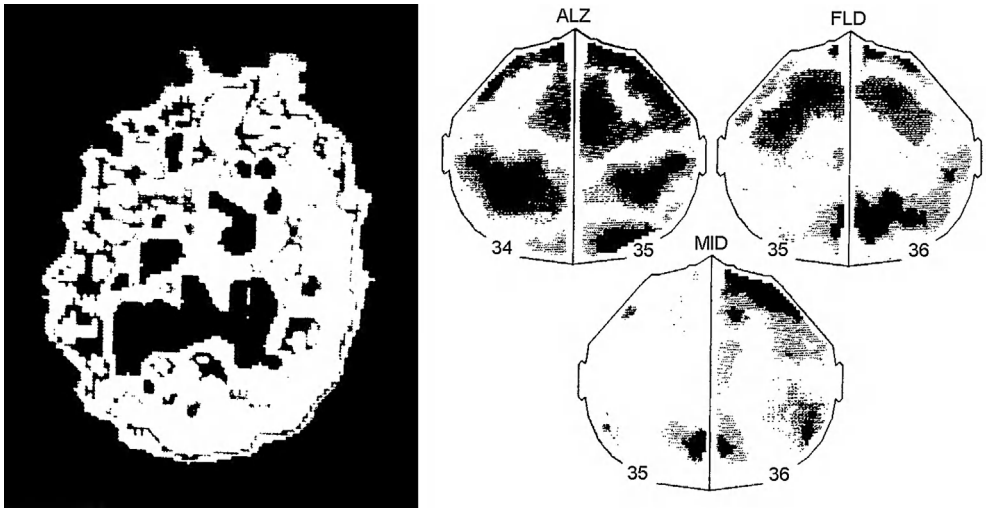


Figure 18.16 (a) A PET scan to identify a blood occlusion, (b) gamma-camera scans of subjects with ALZ, FLD and MID. *(Courtesy Kernforschungsanlage Jülich and Lund Psycho Geriatrics Dept.)*

3) The glucose solution is injected into a patient. Since it is easily metabolized, the glucose goes to the parts of the brain with the highest metabolism, rather than to places with no metabolism.

4) The ^{11}C nuclide decays according to $^{11}C \rightarrow {}^{11}B + e^+$ followed by $e^+ + e^- \rightarrow 2\gamma$. The 0.51 MeV $\gamma\gamma$ coincidences are registered by the PET cameras at various positions.

PET is used especially for studies of brain, heart and lungs. Figure 18.16, left, shows a brain investigation. Because glucose is the only source of energy the brain uses, the rate of glucose metabolism can be assessed throughout the brain, which is an indicator of the brain viability. In this case, the patient has been injected with ^{11}C-methyl glucose and the dark spot indicates an occlusion of the left carotid artery. The technique is quite fast: within a few hours the effect of administered drugs can be revealed.

Although TCT has now superseded other techniques in locating brain tumors, SPECT, gamma-camera and PET have provided dramatically new information of various forms of mental illness, such as epilepsy, manic depression, and dementias such as Alzheimer's disease. The development of new, selective radiopharmaceuticals will not only continue to increase the importance of this diagnostic technique but also contribute to our understanding of the functioning of the normal brain.

The use of positron emitters usually requires fast chemical separation and synthesis techniques. Fast chemical separation techniques for producing pure radioisotopes are described in Chapter 17, but for rapid chemical synthesis techniques, specialized texts should be consulted.

18.5.4. Radiation therapy with internal radionuclides

Radiotracers are also used for therapy though to less extent than in diagnosis (C3b in Table 18.3). The main application is the use of ^{131}I for treatment of thyreotoxicos (Graves disease with enlargement of the thyroid gland), thereby reducing the function of the thyroid. Some data for ^{131}I use: amount administered ca. 200−1000 MBq, organ dose to the thyroid ~340 nGy/Bq (total dose to the glands >500 Sv); whole body dose ~0.04 nGy/Bq. ^{32}P compounds are used in the bloodstream to destroy excess red blood cells in polycythemia. Diphosphonate compounds localize in normal bone tissue, but concentrate in cancerous bone at relatively higher levels (5−15 times higher). Patients receiving this type of radiopharmaceutical experience swift lowering of pain levels.

One of the most fascinating aspects of using radionuclide labelled compounds in medical therapy is to develop monoclonal antibodies, which will seek out particular types of cancer cells and bind to them. If a β-, or α-emitting radionuclide is attached to these antibodies, they will deliver a large absorbed dose to the cancer cells without needlessly irradiating the surrounding tissues. Much research is directed towards this goal of cancer therapy.

18.6. INDUSTRIAL USES OF RADIOTRACERS

Industry has applied radiotracers in a very large variety of ways. More than half of the 500 largest manufacturing concerns in the United States use radioisotopes in the production of metals, chemicals, plastics, pharmaceuticals, paper, rubber, clay and glass products, food, tobacco, textiles, and many other products. Radioisotopes are used to study mixing efficiency, effect of chamber geometry, residence time in reactors, flow rates and patterns in columns and towers for fractionation, absorption, racemization, etc. Some of the many uses are listed in Table 18.6 and a few are described below to reflect the scope and value of the industrial applications of radioisotopes. Quite often the radionuclide used is not isotopic with the system studied.

18.6.1. Mixing

Mixing is an important mechanical operation in many industries. Poor mixing may give an unsatisfactory product and low yield of the operation; unnecessary mixing is a waste of time and energy. By adding a radionuclide to the mixing vessel, or by labelling one of the components, the approach to mixing equilibrium can be followed either by external measurement or taking samples at different time intervals. Among examples of this technique are fluxing of cement, gravel, sand, and water to concrete measured by using irradiated pebbles. The homogeneity of glass melts can be determined by adding $^{24}NaHCO_3$ to the melt; organic compounds of ^{95}Zr ($\beta^-\gamma$ $t_{½}$ 64.02 d) have been used to follow homogenization of oil products. Other uses have been the addition of vitamins to flour, coal powder to rubber, water and gas to oil fields, etc.